

Classical Electrodynamics HW 1

Zachary Tullier¹

¹Student, Physics Department, University of Houston -
Clear Lake , Houston, Texas, U.S

September 16, 2026

1 Textbook Question 1.1

Question :

Use Gauss's theorem [and (1.21) if necessary] to prove the following:

- a) Any excess charge placed on a conductor must lie entirely on its surface. (A conductor by definition contains charges capable of moving freely under the action of applied electric fields.)
- b) A closed, hollow conductor shields its interior from fields due to charges out-side, but does not shield its exterior from the fields due to charges placed inside it.
- c) The electric field at the surface of a conductor is normal to the surface and has a magnitude (σ/ϵ_0), where σ is the charge density per unit area on the surface.

Question 1, Section a) Answer:

Assumptions

- Perfect conductor; therefore charges are perfectly free to move within the conductor
- A conductor by definition contains charges capable of moving freely under the action of applied electric fields
- The charges that are located are in static equilibrium; therefore there is no electric field in the volume where the free charges exist. The charges at the surface cannot leave the surface; therefore there is only electric field at the surface, normal to that surface. Therefore any tangential force from the electric field is zero.
- If the surface charge is positive, the electric field is outward, and vice versa.

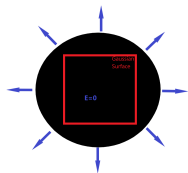


Figure 1: Closed Gaussian Surface within the conductor

Using Gauss's Law

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad (1)$$

Take integral for an arbitrarily shaped 3D solid conductor with excess charge

$$\int_V \nabla \cdot \vec{E} d^3x = \int_V \frac{\rho}{\epsilon_0} dV \quad (2)$$

Using Divergence Theorem

$$\int_V \nabla \cdot A d^3x = \int_S A \cdot n da \quad (3)$$

Substitute equation 3 into equation 2

$$\int_S \vec{E} \cdot n da = \int_V \frac{\rho}{\epsilon_0} dV \quad (4)$$

Evaluate Integral on right

$$\int_S \vec{E} \cdot n da = \sum_i \frac{q_i}{\epsilon_0} \int_S \vec{E} \cdot n da = 0 \quad (5)$$

Physically, we can understand this as the net free charges mutually repelling each other and accelerating away from each other until they hit the surface. The only boundary for mutually repelling, freely moving charges is the surface of the conductor; therefore, they all reside on the surface.

Question 1, Section b) Answer:

Assumptions

- The hollow sphere is connected to ground so that when the external electric field is applied, the charge density of the conductor becomes whatever is opposite of the external electric field.
- Perfect conductor; therefore charges are perfectly free to move within the conductor
- A conductor by definition contains charges capable of moving freely under the action of applied electric fields

First we make a closed Gaussian surface completely within the the hollow conducting shell and place charge outside of the shell

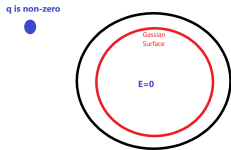


Figure 2: Closed Gaussian Surface within the hollow shell conductor with external electric field

Using Gauss's Law in integral form

$$\oint_S \vec{E} \cdot \vec{n} da = \frac{\int_V \rho_q(x) d^3x}{\epsilon_0} \quad (6)$$

Within the gaussian surface there are no charges ($q = 0$); therefore:

$$\oint_S \vec{E} \cdot n da = 0 \quad (7)$$

If the Gaussian surface was chosen to be mostly along the hollow shells inner surface ($\oint_A \vec{E} \cdot n da$) and the rest of the gaussian surface ($\oint_B \vec{E} \cdot n da$) then we would get

$$\oint_A \vec{E} \cdot n da + \oint_B \vec{E} \cdot n da = 0 \quad (8)$$

Since $\oint_A \vec{E} \cdot n da = 0$ then, $\oint_B \vec{E} \cdot n da = 0$; Therefore, any external field will not have any effect on the inner surface of the conductor, let alone the hollow interior.

Now we consider the same conductor with a charge on the inside

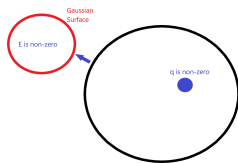


Figure 3: Closed Gaussian Surface within the hollow shell conductor with internal electric field

Using Gauss's Law in integral form (Reference equation 6)

$$\oint_S \vec{E} \cdot n da = \frac{\int_V \rho_q(x) d^3x}{\epsilon_0}$$

The charge on the outer surface of the hollow conducting shell induced by the charge within will generate an electric field such that:

$$\oint_S \vec{E} \cdot n da \neq 0 \tag{9}$$

Question 1, Section c) Answer:

Assumptions

- Perfect conductor; therefore charges are perfectly free to move within the conductor
- A conductor by definition contains charges capable of moving freely under the action of applied electric fields
- The charges that are located are in static equilibrium; therefore there is no electric field in the volume where the free charges exist. The charges at the surface cannot leave the surface; therefore there is only electric field at the surface, normal to that surface. Therefore any tangential force from the electric field is zero.
- If the surface charge is positive, the electric field is outwards, and vice versa.

We consider a hollow conducting box

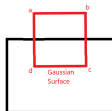


Figure 4: Closed Gaussian Surface covering the inside and outside of a hollow box conductor

Using Gauss's Law in integral form (Reference equation 6)

$$\oint_S \vec{E} \cdot n da = \frac{\int_V \rho_q(x) d^3x}{\epsilon_0}$$

Using points a-d, as shown in Figure 4??:

$$\int_{ab} \vec{E} \cdot n da + \int_{bc} \vec{E} \cdot n da + \int_{cd} \vec{E} \cdot n da + \int_{da} \vec{E} \cdot n da = \quad (10)$$

The two lines that cross the conductor boundary cancel each other out

$$\int_{bc} \vec{E} \cdot n da + \int_{da} \vec{E} \cdot n da = 0 \quad (11)$$

The line within the the conducting box will have no electric field

$$\int_{cd} \vec{E} \cdot n da = 0 \quad (12)$$

Therefore:

$$\int_{ab} \vec{E} \cdot n da = \frac{\int_V \rho_q(x) d^3x}{\epsilon_0} \quad (13)$$

If the only gaussian surface being affected is the top one, then the only charge being considered is the top surface of the conducting sphere. Shrink the conducting box to be "very" small.

$$\vec{E}A = \frac{q}{\epsilon_0} \quad (14)$$

Define Charge per Unit Area (ρ)

$$\rho = \frac{q}{A} \quad (15)$$

Isolate E

$$E = \frac{\rho}{\epsilon_0} \tag{16}$$

2 Textbook Question 1.2

Question:

The Dirac delta function in three dimensions can be taken as the improper limit as $\alpha \rightarrow 0$ of the Gaussian function

$$D(\alpha; x, y, z) = (2\pi)^{-\frac{3}{2}} \alpha^{-3} e^{-\frac{1}{2\alpha^2}(x^2+y^2+z^2)}$$

Consider a general orthogonal coordinate system specified by the surfaces $u = \text{constant}$, $v = \text{constant}$, and $w = \text{constant}$ with length elements $\frac{du}{U}$, $\frac{dv}{V}$, and $\frac{dw}{W}$ in the three perpendicular directions. Show that

$$\delta(x - x') = \delta(u - u')\delta(v - v')\delta(w - w') \cdot UVW$$

by considering the limit of the Gaussian above. Note that as $\alpha \rightarrow 0$ only the infinitesimal length element need be used for the distance between the points in the exponent.

Using the Dirac Delta property

$$\int_{-\text{inf}}^{\text{inf}} \int_{-\text{inf}}^{\text{inf}} \int_{-\text{inf}}^{\text{inf}} \delta(x)\delta(y)\delta(z) dx dy dz = 1 \quad (17)$$

Use the gaussian equation instead of deltas

$$\int_{-\text{inf}}^{\text{inf}} \int_{-\text{inf}}^{\text{inf}} \int_{-\text{inf}}^{\text{inf}} \lim_{\alpha \rightarrow 0} D(\alpha; x, y, z) dx dy dz = 1 \quad (18)$$

Use length elements

$$\int_{-\text{inf}}^{\text{inf}} \int_{-\text{inf}}^{\text{inf}} \int_{-\text{inf}}^{\text{inf}} \lim_{\alpha \rightarrow 0} D(\alpha; x, y, z \rightarrow u, v, w) \frac{du}{U} \frac{dv}{V} \frac{dw}{W} = 1 \quad (19)$$

Compare

$$D(\alpha; x, y, z \rightarrow u, v, w) \frac{du}{U} \frac{dv}{V} \frac{dw}{W} = \delta(x-x') = \delta(u-u')\delta(v-v')\delta(w-w') \quad (20)$$

Isolate D

$$D(\alpha; x, y, z \rightarrow u, v, w) = \delta(x-x') = \delta(u-u')\delta(v-v')\delta(w-w')UVW \quad (21)$$

By definition $D(\alpha; x, y, z \rightarrow u, v, w) = \delta(x-x')$

Therefore;

$\delta(x-x') = \delta(x-x') = \delta(u-u')\delta(v-v')\delta(w-w')UVW$

(22)

3 Textbook Question 3

Question:

Using Dirac delta functions in the appropriate coordinates, express the following charge distributions as three-dimensional charge densities $\rho(x)$

- In spherical coordinates, a charge Q uniformly distributed over a spherical shell of radius R .
- In cylindrical coordinates, a charge A per unit length uniformly distributed over a cylindrical surface of radius b .
- In cylindrical coordinates, a charge Q spread uniformly over a flat circular disc of negligible thickness and radius R .
- The same as part (c), but using spherical coordinates.

Question 3, Section 1 Answer:

For a uniformly charged (Q) shell with radius $r = R$ and a normalization constant (A), the charge density is

$$\rho(x) = A\delta(r - R) \quad (23)$$

Integrate the density over the volume of the shell to get Q

$$Q = \int \int \int \rho(x) dx = \int_0^\pi \int_0^{2\pi} \int_0^{\text{inf}} A\delta(r - R)(r^2 \sin(\theta)) dr d\phi d\theta \quad (24)$$

$$Q = A(2)(2\pi)(R^2)$$

Isolate A

$$A = \frac{Q}{4\pi R^2} \quad (25)$$

Making the charge density

$$\rho(x) = \frac{Q}{4\pi R^2} \delta(r - R) \quad (26)$$

Question 3, Section 2 Answer:

For a uniformly charged (λ) cylindrical surface with radius $r = b$ and a normalization constant (A), the charge density is

$$\rho(x) = A\delta(r - b) \quad (27)$$

Integrate the density over the volume to get λ

$$\lambda = \int \int \int \rho(x) dx = \int_0^1 \int_0^{2\pi} \int_0^{\text{inf}} A\delta(r - b)(r dr d\phi dz) \quad (28)$$

$$Q = A(1)(2\pi)(b)$$

Isolate A

$$A = \frac{\lambda}{2\pi b} \quad (29)$$

Making the charge density

$$\rho(x) = \frac{\lambda}{2\pi b} \delta(r - b) \quad (30)$$

Question 3, Section 3 Answer:

For a uniformly charged (Q) circular disk with radius from $r = 0$ to $r = R$ and a normalization constant (A), the charge density is

$$\rho(x) = A(H(r - 0) - H(r - R))\delta(z) \quad (31)$$

Integrate the density over the volume to get Q

$$Q = \int \int \int \rho(x) dx = \int_{-1}^1 \int_0^{2\pi} \int_0^{\text{inf}} A(H(r-0) - H(r-R))\delta(z)(r dr d\phi dz) \quad (32)$$

$$Q = A(1)(2\pi)\left(\frac{R^2}{2}\right)$$

Isolate A

$$A = \frac{Q}{\pi R^2} \quad (33)$$

Making the charge density

$$\rho(x) = \frac{Q(H(r - 0) - H(r - R))\delta(z)}{\pi R^2}$$

(34)

Question 3, Section 4 Answer:

For a uniformly charged (Q) circular disk with radius from $r = 0$ to $r = R$ and a normalization constant (A), the charge density is

$$\rho(x) = A(H(r - 0) - H(r - R))\delta\left(\theta - \frac{\pi}{2}\right) \quad (35)$$

Integrate the density over the volume to get Q

$$Q = \int \int \int \rho(x) dx \quad (36)$$

$$\begin{aligned} Q &= \int_0^\pi \int_0^{2\pi} \int_0^{\text{inf}} A(H(r-0) - H(r-R))\delta(\theta - \frac{\pi}{2})(r^2 \sin(\theta) dr d\phi d\theta) \\ &= A(1)(2\pi)(\frac{R^3}{3}) \end{aligned}$$

Isolate A

$$A = \frac{3Q}{2\pi R^3} \quad (37)$$

Making the charge density

$$\boxed{\rho(x) = \frac{3Q(H(r-0) - H(r-R))\delta(\theta - \frac{\pi}{2})}{2\pi R^3}} \quad (38)$$

4 Textbook Question 1.4

Question:

Each of three charged spheres of radius a , one conducting, one having a uniform charge density within its volume, and one having a spherically symmetric charge density that varies radially as r^n ($n > |3|$), has a total charge Q . Use Gauss's theorem to obtain the electric fields both inside and outside each sphere. Sketch the behavior of the fields as a function of radius for the first two spheres, and for the third with $n = -2 + 2$.

I hate that the question notes the radius of each sphere to be a when we use R typically. Also, da is area, so that would be confusing. The below formulations will use R as the radius of the sphere.

Question 4, Section 1 Answer:

Using Gauss Law in integral form (Reference equation 6)

$$\oint_S \vec{E} \cdot n da = \frac{\int_V \rho_q(x) d^3x}{\epsilon_0} \quad (39)$$

The sphere is symmetrical, so $\vec{E}$ is constant

$$\vec{E} \oint_S da = \frac{\int_V \rho_q(x) d^3x}{\epsilon_0} \quad (40)$$

If the sphere is "very small" then it acts like a point charge

$$\vec{E} \oint_S da = \frac{Q}{\epsilon_0} \quad (41)$$

Integrate

$$E 4\pi r^2 = \frac{Q}{\epsilon_0} \quad (42)$$

Isolate E to get the electric field outside of the sphere

$$\boxed{E_{outside} = \frac{Q}{4\pi r^2 \epsilon_0}} \quad (43)$$

The electric field is non-existent inside of the closed surface

$$E_{inside} = 0 \quad (44)$$

Question 4, Section 2 Answer:

Using Gauss Law in integral form (Reference equation 6)

$$\oint_S \vec{E} \cdot n da = \frac{\int_V \rho_q(x) d^3x}{\epsilon_0} \quad (45)$$

The sphere is symmetrical, so $\vec{E}$ is constant

$$\vec{E} \oint_S da = \frac{\int_V \rho_q(x) d^3x}{\epsilon_0} \quad (46)$$

The charge density (ρ) takes into account the uniform charge density within the sphere's volume (Q) and the charge on the surface

$$\rho = \left(\frac{Q}{\frac{4}{3}\pi R^3}\right) \left(\frac{4}{3}\pi r^3\right) \quad (47)$$

Reduce algebraically, plug into reduced gauss law, and integrate

$$E 4\pi r^2 = \frac{Q r^3}{\epsilon_0 R^3} \quad (48)$$

Isolate E

$$E = \frac{Qr}{4\pi\epsilon_0 R^3} \quad (49)$$

Since the inside of the conducting sphere is shielded from electric fields outside of the inner wall, we will only consider the charge density in the volume of the sphere.

$$\rho = \left(\frac{Q}{\frac{4}{3}\pi R^3}\right) \quad (50)$$

Reduce algebraically, plug into reduced gauss law, and integrate

$$E4\pi r^2 = \frac{3Qr^3}{4\epsilon_0\pi R^3} \quad (51)$$

Isolate E

$$E = \frac{3Q}{\epsilon_0\pi^2 r^2 R^3} \quad (52)$$

Question 4, Section 2 Answer:

For the outside of the sphere, it has the same electric field as equation [43](#)

$$E_{outside} = \frac{Q}{4\pi r^2 \epsilon_0} \quad (53)$$

For the inside we will again begin by using Gauss Law in integral form (Reference equation [6](#))

$$\oint_S \vec{E} \cdot n da = \frac{\int_V \rho_q(x) d^3x}{\epsilon_0} \quad (54)$$

The sphere is symmetrical, so $\vec{E}$ is constant (Reference equation 46)

$$\vec{E} \oint_S da = \frac{\int_V \rho_q(x) d^3x}{\epsilon_0} \quad (55)$$

The charge density is not constant, but varies as r^n .

$$\rho = Ar^n \quad (56)$$

Integrate the charge density over the whole sphere to find A , the constant

$$\int_0^{2\pi} \int_0^\pi \int_0^R Ar^n r^2 \sin(\theta) d\theta d\phi dr = Q \quad (57)$$

Isolate A

$$A = \frac{Q}{4\pi \left(\frac{R^{n+3}}{n+3}\right)} \quad (58)$$

Plug into gauss and convert coordinates to polar

$$\vec{E} \oint_S da = \frac{\int_0^{2\pi} \int_0^\pi \int_0^R \frac{Q}{4\pi \left(\frac{R^{n+3}}{n+3}\right)} r^n r^2 \sin(\theta) d\theta d\phi dr}{\epsilon_0} \quad (59)$$

Integrate

$$E 4\pi r^2 = \frac{Q r^{n+3}}{R^{n+3} \epsilon_0} \quad (60)$$

Isolate E

$$E = \frac{Q r^{n+3}}{4\pi \epsilon_0 r^2 R^{n+3}} \quad (61)$$

for $n = -2$

$$E = \frac{Q}{4\pi\epsilon_0 r R} \tag{62}$$

for $n = +2$

$$E = \frac{Qr^3}{4\pi\epsilon_0 R^5} \tag{63}$$

5 Textbook Question 5

Question:

The time-averaged potential of a neutral hydrogen atom is given by

$$\Phi = \frac{q}{4\pi\epsilon_0} \frac{e^{\alpha r}}{r} \left(1 + \frac{\alpha r}{2}\right) \quad (64)$$

where q is the magnitude of the electronic charge, and $\alpha^{-1} = \frac{a_0}{2}$, a_0 being the Bohr radius. Find the distribution of charge (both continuous and discrete) that will give this potential and interpret your result physically.

Question 5, Section 1 Answer:

From Poisson's equation and gauss's law in integration form

$$\nabla^2 \Phi = -\frac{\rho}{\epsilon_0} \quad (65)$$

Expand the gradient into its individual spherical coordinates

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Phi}{\partial r} \right) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial \Phi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2 \Phi}{\partial \phi^2} = -\frac{\rho}{\epsilon_0} \quad (66)$$

The potential is spherically symmetric, so $\frac{\partial}{\partial \theta} = 0$ and $\frac{\partial^2 \Phi}{\partial \phi^2} = 0$
The reduced poisson equation is

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Phi}{\partial r} \right) = -\frac{\rho}{\epsilon_0} \quad (67)$$

Insert given formula for Φ

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \left(\frac{q}{4\pi\epsilon_0} \frac{e^{\alpha r}}{r} \left(1 + \frac{\alpha r}{2}\right) \right)}{\partial r} \right) = -\frac{\rho}{\epsilon_0} \quad (68)$$

Reduce and isolate q

$$\rho = \frac{-q\alpha^3 e^{-\alpha r}}{8\pi} - \frac{e^{-\alpha r} q}{4\pi r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \frac{1}{r} \right) \quad (69)$$

If $r = 0$ then this equation contains singularities, infinities, and is non-physical. We will consider the same equation for two cases ($r \approx 0$ and $r > 0$) For $r \approx 0$, $e^{\alpha r} \approx 1$ so

$$\rho = \frac{-q\alpha^3}{8\pi} - \frac{q}{r\pi} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \frac{1}{r} \right) \quad (70)$$

Using the delta relation

$$\nabla^2 \frac{1}{r} = -4\pi\delta(r) \quad (71)$$

Expand the gradient into its individual spherical coordinates

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \frac{1}{r} \right) = -4\pi\delta(r) \quad (72)$$

Substitute equation 72 in equation 70

$$\rho = \frac{-q\alpha^3}{8\pi} - \frac{q}{r\pi} (-4\pi\delta(r)) \quad (73)$$

Reduce

$$\rho = \frac{-q\alpha^3}{8\pi} + q\delta(r) \quad (74)$$

For $r > 0$ evaluate derivatives normally

$$\rho = \frac{-q\alpha^3}{8\pi} e^{-\alpha r} \quad (75)$$

The equations can be combined because if $r \approx 0$ this general equation reduces to it "case" equation

$$\boxed{\rho = \frac{-q\alpha^3}{8\pi}e^{-\alpha r} + q\delta(r)} \quad (76)$$

6 Custom Notes Question

Question:

Separation of Variables: Laplace Equation in Spherical Coordinates

We begin with the laplace equation

$$\nabla^2 \Phi = 0 \quad (77)$$

Expanding the nabla to spherical polar coordinates

$$\frac{1}{r} \frac{\partial^2}{\partial r^2} (r\Phi) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} (\sin(\theta) \frac{\partial \Phi}{\partial \theta}) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2 \Phi}{\partial \phi^2} = 0 \quad (78)$$

Use the method of separation by trying a solution of the form

$$\Phi(r, \theta, \phi) = \frac{R(r)}{r} P(\theta) Q(\phi) \quad (79)$$

Substitute equation 79 into equation 78

$$\begin{aligned} & \frac{1}{r} \frac{\partial^2}{\partial r^2} \left(r \frac{R(r)}{r} P(\theta) Q(\phi) \right) \\ & + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial \frac{R(r)}{r} P(\theta) Q(\phi)}{\partial \theta} \right) \\ & + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2 \frac{R(r)}{r} P(\theta) Q(\phi)}{\partial \phi^2} = 0 \end{aligned}$$

Reduce algebraically

$$\begin{aligned}
& P(\theta)Q(\phi)\frac{1}{r}\frac{\partial^2 R(r)}{\partial r^2} \\
& + \frac{R(r)}{r}Q(\phi)\frac{1}{r^2\sin(\theta)}\frac{\partial}{\partial\theta}(\sin(\theta)\frac{\partial P(\theta)}{\partial\theta}) \\
& + \frac{R(r)}{r}P(\theta)\frac{1}{r^2\sin^2(\theta)}\frac{\partial^2 Q(\phi)}{\partial\phi^2} = 0
\end{aligned}$$

Reduce by multiplying the entire equation by $\frac{r^3\sin^2(\theta)}{R(r)P(\theta)Q(\phi)}$

$$\frac{r^2\sin^2(\theta)}{R(r)}\frac{d^2 R(r)}{dr^2} + \frac{\sin(\theta)}{P(\theta)}\frac{d}{d\theta}(\sin(\theta)\frac{dP(\theta)}{d\theta}) + \frac{1}{Q(\phi)}\frac{d^2 Q(\phi)}{d\phi^2} = 0 \quad (80)$$

A singular term is only dependent on 1 coordinate

$$-m^2 = \frac{1}{Q(\phi)}\frac{d^2 Q(\phi)}{d\phi^2} \quad (81)$$

Rearrange equation to be in differential form

$$\frac{d^2 Q(\phi)}{d\phi^2} = -m^2 Q(\phi) \quad (82)$$

General solution for $Q(\phi)$

$$Q(\phi) = A_m e^{im\phi} + B_m e^{-im\phi} \quad (83)$$

Substitute equation 81 into equation 80

$$\frac{r^2\sin^2(\theta)}{R(r)}\frac{d^2 R(r)}{dr^2} + \frac{\sin(\theta)}{P(\theta)}\frac{d}{d\theta}(\sin(\theta)\frac{dP(\theta)}{d\theta}) - m^2 = 0 \quad (84)$$

Reduce by multiplying the entire equation by $\sin^2(\theta)$

$$\frac{r^2}{R(r)} \frac{d^2 R(r)}{dr^2} + \frac{1}{\sin(\theta)P(\theta)} \frac{d}{d\theta}(\sin(\theta)) \frac{dP(\theta)}{d\theta} - \frac{m^2}{\sin^2(\theta)} = 0 \quad (85)$$

Two terms are dependent on 1 coordinate

$$-l(l+1) = \frac{1}{\sin(\theta)P(\theta)} \frac{d}{d\theta}(\sin(\theta)) \frac{dP(\theta)}{d\theta} - \frac{m^2}{\sin^2(\theta)} \quad (86)$$

Substitute equation 86 into equation 85

$$\frac{r^2}{R(r)} \frac{d^2 R(r)}{dr^2} - l(l+1) = 0 \quad (87)$$

Rearrange equation to be in differential form

$$\frac{d^2 R(r)}{dr^2} = -l(l+1) \frac{R(r)}{r^2} \quad (88)$$

General solution for $R(r)$

$$R(r) = A_l r^{l+1} + B_l r^{-l} \quad (89)$$